Exponentials & Polynomials

ANSWER KEY

Section 1: Laws of Exponents & Radicals

1 If $x > 0$, which expression is equivalent to $\sqrt[3]{x^6 \cdot x^9}$?

a. x^5 ✓

b. x^3

c. x^{15}

d. x^6

Combine exponents: $x^6 \cdot x^9 = x^{15}$. Then $\sqrt[3]{x^{15}} = x^{15/3} = x^5$.

2 If $2^{3x-1} = 32$, what is the value of x?

a. 1

b. 2 ✓

c. 3

d. 4

Rewrite 32 as 2^5: $2^{3x-1} = 2^5 \Rightarrow 3x - 1 = 5 \Rightarrow 3x = 6 \Rightarrow x = 2$.

3 Which of the following is equivalent to $\frac{x^{5/2}}{x^{1/2}}$ for all $x > 0$?

a. x^2 ✓

b. x^3

c. x^5

d. $x^{5/4}$

Subtract exponents: $x^{5/2 - 1/2} = x^{4/2} = x^2$.

Section 2: Exponential Growth and Decay Models

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4 A town had a population of $12,000$ in the year 2010 and increases by 3.5% each year. Which function models the population $P(t)$, where t is the number of years since 2010?

a. $P(t) = 12000(1.35)^t$

b. $P(t) = 12000(1.035)^t$ ✓

c. $P(t) = 12000(0.965)^t$

d. $P(t) = 12000 + 35t$

Growth formula is $P(t) = P_0(1 + r)^t$. Here $r = 0.035$, so the multiplier is $1 + 0.035 = 1.035$. Thus $P(t) = 12000(1.035)^t$.

5 A certain medication decays in the bloodstream according to $A(t) = 500(0.75)^t$, where $A(t)$ is in milligrams and t is in hours. By what percentage does the amount of medication decrease each hour?

a. 75%

b. 25% ✓

c. 50%

d. 15%

The base is $b = 1 - r = 0.75 \Rightarrow r = 0.25$, which represents a 25% hourly decrease.

6 An investment account triples in value every 8 years. If the initial balance is 1,500 dollars, which expression represents the account balance after t years?

a. $1500(3)^{t/8}$ ✓

b. $1500(3)^{8t}$

c. $1500(8)^{t/3}$

d. $1500(1.03)^{8t}$

When an amount triples every k years, the model is $A(t) = A_0 \cdot 3^{t/k}$. Here $k = 8$, yielding $1500(3)^{t/8}$.

Section 3: Polynomial Operations & Factoring

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7 Which expression is the quotient when $2x^3 - 5x^2 + 7x - 4$ is divided by $x - 1$?

a. $2x^2 - 3x + 4$ ✓

b. $2x^2 + 3x - 4$

c. $2x^2 - 7x + 4$

d. $2x^2 - 3x - 4$

Use polynomial division or synthetic division with $c = 1$: $1|2 \quad -5 \quad 7 \quad -4 \quad \downarrow \quad 2 \quad -3 \quad 4 \quad 2 \quad -3 \quad 4 \quad 0$. The quotient is $2x^2 - 3x + 4$ with remainder 0.

8 If polynomial $P(x) = x^3 - kx^2 + 4x - 12$ has a zero at $x = 3$, what is the value of k?

a. 2

b. 3 ✓

c. 4

d. 5

By the Factor Theorem,
$P(3) = 0 \Rightarrow 3^3 - k(3^2) + 4(3) - 12 = 0 \Rightarrow 27 - 9k + 12 - 12 = 0 \Rightarrow 27 - 9k = 0 \Rightarrow k = 3$.

9 Which of the following is equivalent to $(3x - 2)(x^2 + 4x - 5)$?

a. $3x^3 + 10x^2 - 23x + 10$ ✓

b. $3x^3 + 12x^2 - 15x + 10$

c. $3x^3 - 2x^2 + 8x - 10$

d. $3x^3 + 10x^2 - 15x - 10$

Expand:
$3x(x^2 + 4x - 5) - 2(x^2 + 4x - 5) = 3x^3 + 12x^2 - 15x - 2x^2 - 8x + 10 = 3x^3 + 10x^2 - 23x + 10$.

Section 4: Polynomial Graphs & Roots

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10 A polynomial function $f(x)$ has zeros at $x = -2, 1, 4$. Which equation could define f?

a. $f(x) = (x + 2)(x - 1)(x - 4)$ ✓

b. $f(x) = (x - 2)(x + 1)(x + 4)$

c. $f(x) = (x + 2)^2(x - 1)$

d. $f(x) = x^3 - 2x + 4$

A polynomial with roots r_1, r_2, r_3 factors as $a(x - r_1)(x - r_2)(x - r_3)$. Substituting $-2, 1, 4$ gives $(x + 2)(x - 1)(x - 4)$.

11 The graph of $y = (x - 3)^2(x + 1)$ touches the x-axis without crossing it at which value of x?

a. $x = -1$

b. $x = 3$ ✓

c. $x = 0$

d. $x = 1$

A factor with an even multiplicity, such as $(x - 3)^2$, causes the graph to be tangent to the x-axis (touch without crossing) at $x = 3$.

12 What is the degree of the polynomial function $g(x) = (x^2 - 1)(2x - 3)^3$?

a. 4

b. 5 ✓

c. 6

d. 7

The degree of a product is the sum of the degrees: degree of $(x^2 - 1)$ is 2, degree of $(2x - 3)^3$ is 3. Total degree $= 2 + 3 = 5$.

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